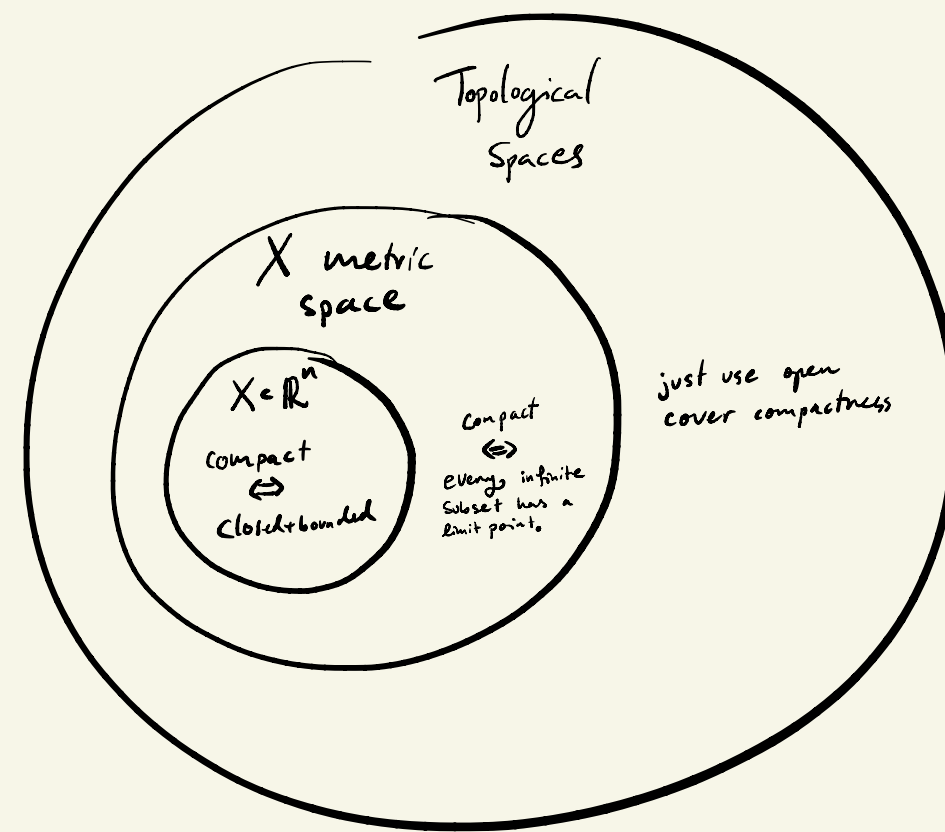


Topics

- Compact $\Rightarrow$ closed
- Closed subsets of compact set are cpct.
- Nested intersections of cpct. are nonempty
- Infinite subset of cpct. has limit pt.
- The d-cell is compact
- Compact in $\mathbb{R}^d \Leftrightarrow$ closed and bounded



(Heine-Borel theorem)

Thm (2.34) Compact subsets of metric spaces are closed

Proof Let $K \subset X$ be compact.

Let $p \in K^c$.

For all $q \in K$, let

$$V_q = N_{\frac{d(p,q)}{2}}(q)$$

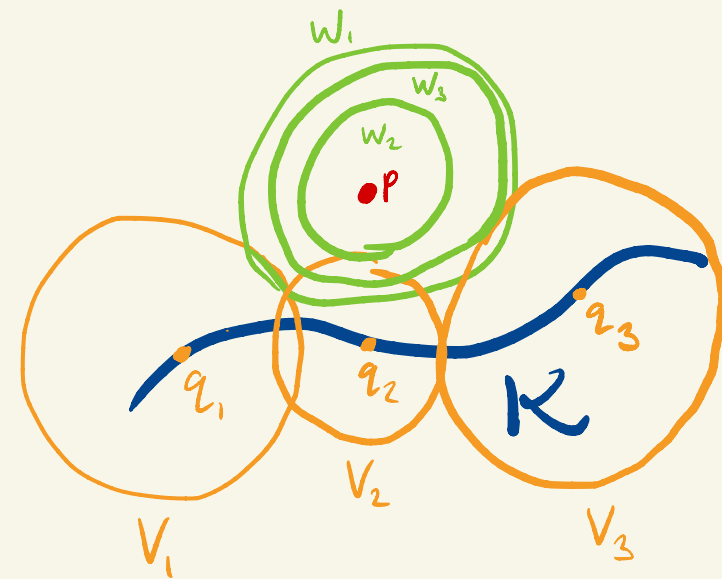
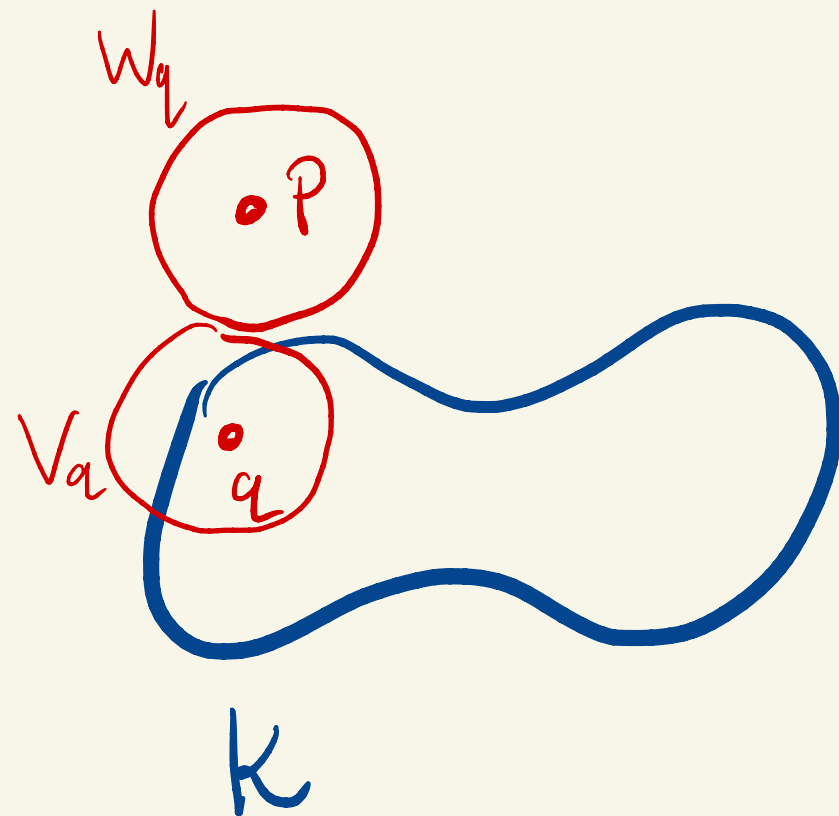
$$W_q = N_{\frac{d(p,q)}{2}}(p)$$

$\{W_q\}_{q \in K}$ is an open cover of K .

There is a finite set of pts $q_1, \dots, q_k$ s.t. $K \subset V_{q_1} \cup \dots \cup V_{q_k}$.

$W_{q_1} \cap \dots \cap W_{q_k}$ is a nbhd of p disjoint from K .

K^c is open, so K is closed.



Thm (2.35) Closed subsets of compact sets are compact.

proof:

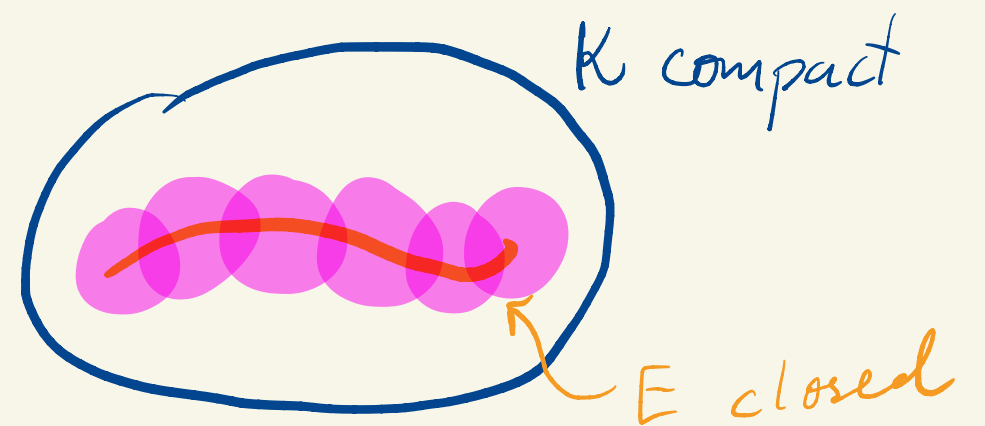
Suppose $E \subset K \subset X$, E closed, K compact.

Let $\{G_\alpha\}$ be a cover of E .

$\{G_\alpha\} \cup \{E^c\}$ is an open cover of K .

$\exists$ finite subcover $G_{\alpha_1}, \dots, G_{\alpha_k}, E^c$ of K .

$G_{\alpha_1}, \dots, G_{\alpha_k}$ is a cover of E .



Corollary F closed, K compact $\Rightarrow F \cap K$ compact.

proof: K closed $\Rightarrow F \cap K$ is closed

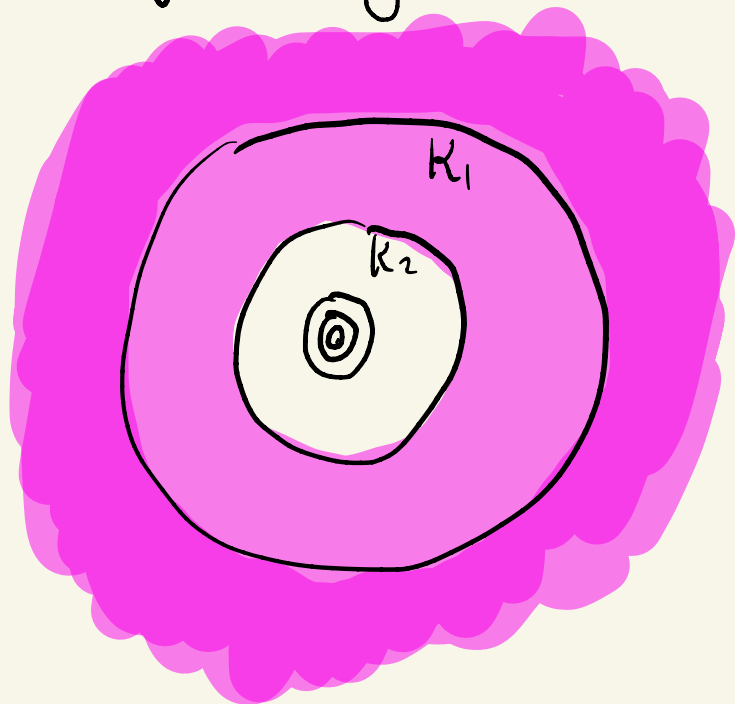
$F \cap K$ closed, K cpt. $\Rightarrow F \cap K$ cpt.

Theorem (2.36; special case) If $K_1 \supset K_2 \supset K_3 \supset \dots$ are cpt., nonempty then $\bigcap_{n \in \mathbb{N}} K_n \neq \emptyset$.

Finite intersection in a nested seq. gives smallest set.

Proof Suppose $\bigcap_{n \in \mathbb{N}} K_n = \emptyset$. Then $\bigcup K_n^c = X$, so $\{K_n^c\}$ is an open cover of X , thus also of K_1 . Let $\{K_{n_1}^c, \dots, K_{n_k}^c\}$ be a finite subcover of K_1 .

$\bigcup_{i=1}^k K_{n_i}^c = K_{\max\{n_i\}}^c$ by nesting, so K_1 is disjoint from $K_{\max\{n_i\}}$, contradiction.

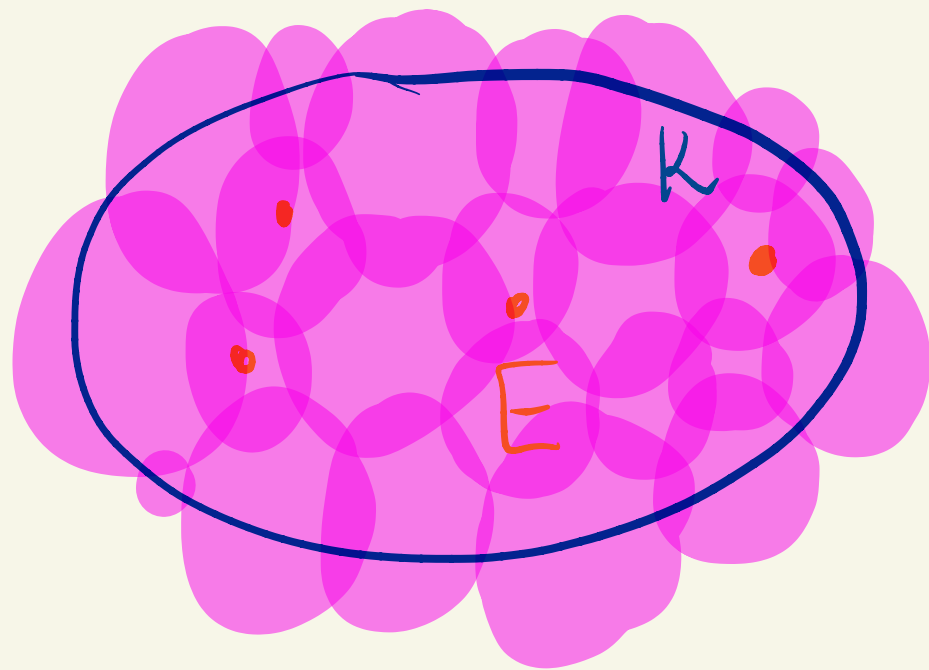


Thm (1.37) If E is an infinite subset of a compact set K ,
then E has a limit point.

proof: Contrapositive.

Suppose E has no limit pts. Let V_q be nbhd of $q \in K$
s.t. $|V_q \cap E| \leq 1$. $\{V_p\}$ is an open cover of K , so

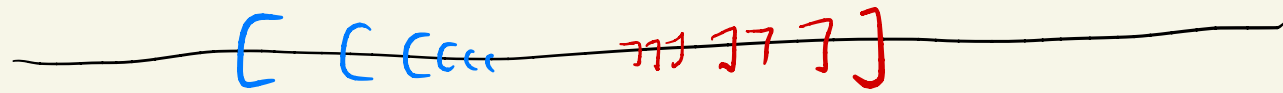
has a finite subcover $V_{p_1}, \dots, V_{p_k}$.



So $|E| \leq k$, i.e. E is finite.

Theorem Suppose $[a_1, b_1] \supset [a_2, b_2] \supset [a_3, b_3] \supset \dots$ is a nested seq. of non-empty closed intervals.

Then $\bigcap_{n \in \mathbb{N}} [a_n, b_n] \neq \emptyset$.



Proof: $A = \{a_i \mid i \in \mathbb{N}\}$ is non-empty and bdd above.

We show $a = \sup A \in \bigcap_{n \in \mathbb{N}} [a_n, b_n]$.

① a is upper bound for A so $a \geq a_i \forall i \in \mathbb{N}$.

② b_k is upper bound of A for all k (check), so $a \leq b_k \forall k$

$\Rightarrow a \in [a_k, b_k] \forall k$

Def A d-cell in $\mathbb{R}^d$ is a set of the form

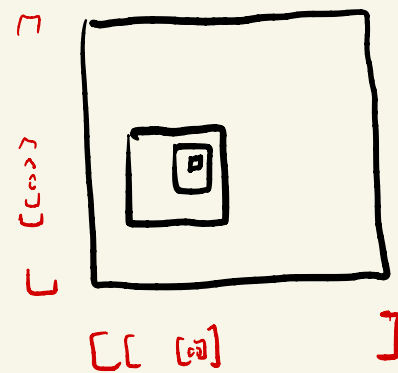
$$[a_1, b_1] \times \cdots \times [a_d, b_d] =$$

Thm If $I_1 \subset I_2 \subset \cdots$ is a nested sequence of d-cells in $\mathbb{R}^d$,
 then $\bigcap_{n \in \mathbb{N}} I_n \neq \emptyset$.

Suppose $I_n = [a_{1n}, b_{1n}] \times \cdots \times [a_{dn}, b_{dn}]$

have $x_1, \dots, x_d$ s.t. $x_i \in [a_{in}, b_{in}] \quad \forall n, i$.

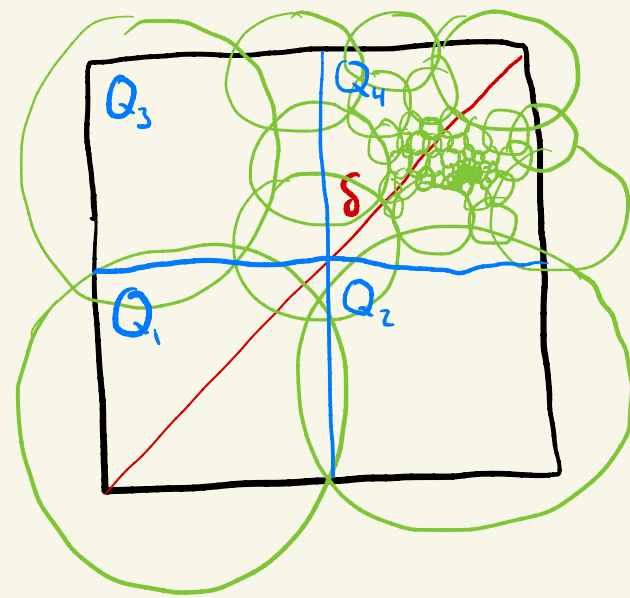
$\Rightarrow X = (x_1, \dots, x_d) \in I_n \quad \forall n$



Thm (2.40) Every d -cell is compact

Let $I = [a_1, b_1] \times \dots \times [a_d, b_d]$

Let $\delta = \sqrt{\sum (a_j - b_j)^2}$ be the diameter of I



Suppose $\{Q_k\}$ covers I with no finite subcover.

Let $c_j = \frac{a_j + b_j}{2}$

$I = \bigcup_{k=1}^{\infty} Q_k$ where Q_k are products of $[a_j, c_j]$, and $[c_j, b_j]$.

There must be a Q_k which doesn't have a finite subcover. Let $I_1 = Q_k$.

Inductively, we get $I \supset I_1 \supset I_2 \supset \dots$

Let $x \in \bigcap_{k=1}^{\infty} I_k$ using thm 2.39.

Let $x \in N_r(x) \subset G_\alpha$ where G_α is an element of the cover.

I_k has diameter $2^{-k} \delta$, so $I_k \subset N_r(x)$

for sufficiently large k .

This contradicts I_k having no finite subcover.

